World Models for Autonomous Driving: From Future Generation to Decision Making

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***Abstract*—Autonomous driving systems must reason beyond the current scene, which need to predict how surrounding traffic may evolve, assess the consequences of candidate actions, and support safe closed-loop behavior. Recent work has moved from perception and short-horizon prediction toward driving world models that can generate future scenes, roll out possible states, or expose transitions for decision making. Instead of organizing the field simply by modality, architecture, or generative quality, this survey adopts a Model Predictive Control (MPC) perspective, which views driving world models as predictive transition interfaces for state rollout, constraint reasoning, cost evaluation, uncertainty handling, and receding-horizon decision making. Specifically, we group existing word model methods for automous driving into future world generation, planning with world models, and hybrid prediction-planning world models. In addition, we compare visual, bird's-eye-view, occupancy-centric, LiDAR, and latent representations, and we discuss diffusion, autoregressive, latent-dynamics, and hybrid mechanisms. In addition, we also reviewed datasets, simulators, metrics, benchmark protocols, and the evidence gap between open-loop and closed-loop testing. Open problems include long-horizon reliability, unified multimodal state representation, generation-to-planning transfer, efficiency, scalability, and foundation-model-assisted safety evaluation. Overall, driving world models should be judged by whether generated futures improve MPC-style planning, evaluation, and system-level validation, rather than by realism alone. More information is provided in Project Page.**

***Index Terms*—Autonomous driving, world models, model predictive control, future generation, planning, simulation, closed-loop evaluation, intelligent transportation systems.**

I. Introduction

Autonomous driving requires more than accurate recognition of the current traffic scene that must simultaneously anticipate how the environment can change under different actions, interactions, and constraints. We use the term driving world model for a learned model that represents the driving environment, rolls it forward in time, and exposes future states for generation, simulation, evaluation, or decision making. The usefulness of a driving world model extends beyond next-frame or trajectory prediction. Some models generate plausible future scenes from observations, actions, maps, language, or scene structure; others estimate the consequences of candidate behaviors. In a planning loop, a world model can also provide cost-relevant states, rewards, latent transitions, or simulated feedback for closed-loop control. These roles distinguish world models from perception modules, which recover the present state, and from conventional motion prediction modules, which often focus on agent-level trajectories.

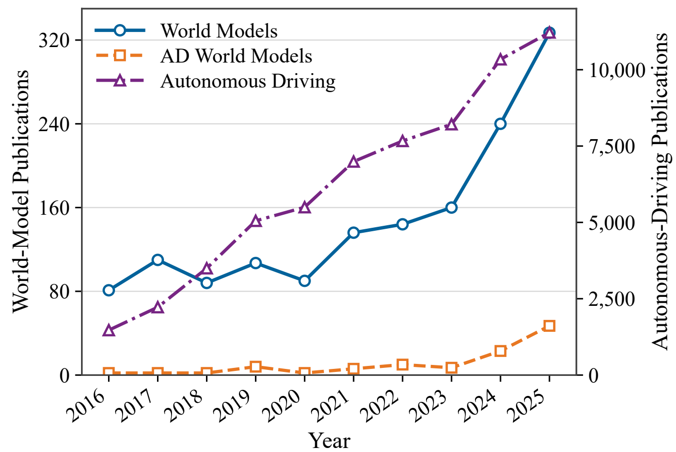

Fig. 1. Annual publication trends of autonomous-driving research, world-model research, and autonomous-driving world-model research from 2016 to 2025. World-model and autonomous-driving world-model counts are plotted against the left axis, while autonomous-driving counts are plotted against the right axis to accommodate the different scales. The statistics were collected from the Web of Science Core Collection using topic-based queries.

World models for autonomous driving have grown rapidly in recent years, with a wide variety of methods, representations, and evaluation protocols. Publication trends justify a focused review. Fig. 1 jointly compares autonomous-driving research, world-model research, and autonomous-driving world-model research from 2016 to 2025. The figure shows steady growth in autonomous-driving publications over the past decade and also shows world models moving from a general machine-learning methodology into an application-specific autonomous-driving topic as attention turns to interactive rollout, closed-loop validation, and decision-aware generation. A dedicated survey can clarify how these methods differ, what they support in the driving pipeline, and where the evidence remains incomplete.

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TABLE I
RECENT SURVEY PAPERS AND THEIR LIMITATIONS RELATIVE TO AUTONOMOUS-DRIVING WORLD MODELS

Study	Year	Focus Area	Key Contributions	Limitations Relative to This Survey
General world-model surveys [1]	2024	General WMs	WM concepts, agents, and cross-domain applications.	Limited AD taxonomy; little planning-interface or closed-loop analysis.
Embodied-AI WM survey [3]	2025	Embodied WMs	Function, time, space, and embodied-agent uses.	Cross-domain focus; driving constraints and traffic interaction are secondary.
Initial AD-WM survey [4]	2024	AD WMs	Early AD-WM concepts and application categories.	Limited task-role taxonomy, planning utility, and benchmark ecology.
AD-WM survey [5]	2025	Driving WMs	Broad roadmap over generation, planning, interaction, and components.	Less explicit MPC-style transition-interface framing.
DWM ecosystem survey [6]	2025	DWM ecosystem	Modalities, applications, datasets, and ecosystem roles.	Mainly modality/ecosystem centered; weaker generation-to-decision focus.
Video-generation WM survey [7]	2024	Video generation	Links video generation with controllable future synthesis.	Emphasizes realism; planner consumption and constraints are less central.
Future physical world generation survey [8]	2025	Future generation	Physical future and scene generation across embodied domains.	Focuses on generation quality, not decision-oriented state exposure.
CAV WM survey [9]	2025	Connected vehicles	Cooperative and infrastructure-assisted WM applications.	Prioritizes CAV settings; general AD-WM planning taxonomy is narrower.
Occupancy perception survey [10]	2024	Occupancy perception	Occupancy representation, fusion, and supervision.	Representation-adjacent; not centered on rollout or planner-facing evidence.
FM scenario survey [11]	2025	FM scenarios	Foundation-model scenario generation, analysis, and testing.	Strong on testing; less focused on learned transition models for MPC.
Planning and MPC surveys [12], [13]	2016/2021	Planning/control	Constraints, vehicle dynamics, optimization, and MPC.	Control foundation only; no learned AD-WM taxonomy or generative models.
Our survey	**2026**	**AD world models**	**Task roles, representations, mechanisms, evaluation, and MPC interfaces.**	**Connects generated states to rollout, cost, constraints, uncertainty, and replanning.**

The literature has grown quickly but remains fragmented. Methods differ in representation space, including image/video, bird's-eye-view (BEV), occupancy, point clouds, and latent states, and in mechanism, including diffusion, autoregressive transformers, recurrent state-space models, and hybrid prediction-planning designs. The larger difference lies in where a world model sits in the driving pipeline: some methods generate futures, some evaluate candidate trajectories, and others support closed-loop simulation or unified decision systems. We make this shift from generation to decision support explicit and treat autonomous-driving world models as a task-level, planning-aware research area.

A. Comparison to Existing Surveys

Existing surveys cover general and embodied world models, autonomous-driving world models, video and future-scene generation, CAV world models, occupancy reasoning, foundation models, and scenario generation [1]–[11]. Motion-planning and MPC surveys examine planning representations, constraint handling, receding-horizon optimization, and autonomous ground-vehicle control [12], [13]. The gap is the missing interface between learned world models and control. World-model surveys often emphasize modality or generative architecture, whereas MPC surveys usually assume an explicit predictive model. We therefore organize autonomous-driving-specific world models by task role, using state prediction, action-conditioned rollout, cost evaluation, constraints, uncertainty, and closed-loop replanning as the MPC reference. Table I summarizes how this position differs from existing survey lines.

The task-role view starts from a practical observation: the same predicted future can be useful, irrelevant, or misleading depending on how a planner consumes it. From an MPC perspective, prediction matters when a planner can query the model under candidate ego actions, constrain the result by road geometry and vehicle dynamics, convert it into cost or risk, and reuse it over a receding horizon. Future video may look convincing yet remain too expensive or weakly grounded for online planning; BEV or occupancy rollout may look plain but expose collision geometry; compact latent transitions may help policy improvement while hiding safety-relevant failures. We therefore treat representation, mechanism, and downstream interface as coupled design choices.

Another motivation is the gap between open-loop prediction quality and closed-loop driving evidence. Large datasets make accurate predictors and generators possible, but planning performance depends on feedback, interaction, and distribution shift after the ego vehicle acts. Recent benchmarks and simulators therefore emphasize closed-loop execution, reactive agents, scenario diversity, and actionability rather than only reconstruction or displacement error [14]–[18]. Driving world models are useful only if evidence connects the generated state to a decision consequence.

B. Distinction, Contribution and Roadmap

Compared with modality-centered surveys, our taxonomy separates generating a future from using a future. Future generation models optimize controllable synthesis, temporal realism, or sensor consistency; planning-oriented models need rewards, values, costs, or risk estimates that can be queried repeatedly. The taxonomy also separates offline prediction from closed-loop world simulation, where the future changes after the ego policy acts. These distinctions matter as world models move from demonstrations toward safety-oriented validation.

The contributions of this survey is summarized as follows:

- We propose a task-oriented taxonomy for autonomous driving world models that groups existing work into future world generation, planning method, and hybrid prediction-planning frameworks.
- We compare representation spaces and modeling mechanisms from an MPC-informed view, focusing on whether a world model exposes state, uncertainty, constraints, and cost-relevant variables for downstream planning.
- We summarize datasets, simulators, metrics, benchmark protocols, and open-loop versus closed-loop evidence, then discuss challenges that affect whether world models can become reliable driving-system components.
- We identify open problems such as long-horizon reliability, unified multimodal state representation, generation-to-planning transfer, efficiency, scalability, and foundation-model-assisted safety evaluation.

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Fig. 2 outlines the roadmap of this survey, linking the definition, taxonomy, comparison, evaluation ecology, and future challenges to the MPC-oriented view of autonomous-driving world models. The remainder of the paper proceeds as follows. Section II defines the scope of driving world models. Section III presents the taxonomy. Section IV compares mechanisms, representations, and utility. Section V reviews datasets, simulators, metrics, and benchmarks. Section VI discusses open challenges, and Section VII concludes this survey.

II. What Is the World Model for Auto-Driving?

A. Definition

We define the world model as a learned, temporally evolving representation of the driving environment that can be conditioned, rolled forward, and consumed by simulation or decision-making modules. Its defining property is not generic prediction, but an explicit organization of world state and state evolution. The state may be a future video, a BEV map, a 3-D occupancy field, a point cloud sequence, or a latent state with decision meaning. A world model relies on an identifiable environment representation, captures temporal evolution rather than only a static state, and can be conditioned on observations, ego actions, planned trajectories, maps, language instructions, or scene metadata. Its outputs should be usable by downstream tasks such as simulation, trajectory evaluation, planning, or safety validation. Short-horizon predictors without rollout ability and policies without explicit world representations fall outside our scope.

The definition separates driving world models from generic generative models. Visually plausible traffic scenes are insufficient unless the model preserves driving constraints such as road topology, kinematic feasibility, agent interaction, occlusion, free space, and the causal effect of ego actions. Conversely, a useful driving world model need not render photorealistic images. A BEV latent, an occupancy sequence, or a value-predictive state can qualify if it supports controlled rollout and decision-relevant evaluation.

For comparison, we describe each driving world model through a learned transition interface. The interface receives a state description and conditions, advances the state in time, and exposes predicted consequences. The state may be explicit, as in occupancy grids, trajectories, or point clouds, or implicit, as in recurrent latent states. Conditions may include ego actions, routes, maps, textual instructions, planned trajectories, or scenario edits. Predicted consequences may include future sensor observations, semantic scenes, collision probabilities, rewards, or distributions over possible futures. The interface makes otherwise different methods comparable.

B. Formal MPC-Oriented Interface

To specify the interface, let $\mathbf{s}_t$ denote a world state at time t, $\mathbf{a}_t$ an ego action or planned motion primitive, and $\mathbf{z}_t$ exogenous context such as map, route, agent history, sensor observation, or language condition. A generic driving world model can then be written as a stochastic transition model:

$$\begin{aligned} \mathbf{s}_{t+1} &\sim \hat{p}_\theta(\cdot \mid \mathbf{s}_t, \mathbf{a}_t, \mathbf{z}_t), \\ \mathbf{o}_{t+1} &\sim \hat{p}_\theta(\cdot \mid \mathbf{s}_{t+1}, \mathbf{z}_{t+1}), \end{aligned} \tag{1}$$

where $\mathbf{o}_{t+1}$ is an optional observation-level output, such as an image, BEV state, occupancy field, LiDAR frame, or latent readout. The notation is deliberately general. Deterministic predictors correspond to a degenerate distribution, whereas diffusion, autoregressive, ensemble, and latent-dynamics models instantiate different choices of $\hat{p}_\theta$.

The transition-interface view connects naturally to model predictive control. Given a horizon H, and using relative horizon indices for compactness, a planner can evaluate candidate action sequences with a receding-horizon objective:

$$\begin{aligned} \mathbf{a}^\star_{0:H-1} = \arg\min_{\mathbf{a}_{0:H-1}} \mathbb{E}_{\hat{p}_\theta}\left[\sum_{k=0}^{H-1} \ell(\mathbf{s}_k, \mathbf{a}_k) + \ell_T(\mathbf{s}_H)\right], \\ \text{s.t. } \mathbf{s}_{k+1} \sim \hat{p}_\theta(\cdot \mid \mathbf{s}_k, \mathbf{a}_k, \mathbf{z}_k), \\ h(\mathbf{s}_k, \mathbf{a}_k) \le 0, \quad k = 0, \dots, H-1, \end{aligned} \tag{2}$$

where ℓ and ℓ_T denote running and terminal costs, while h represents feasibility or safety constraints. Equation (2) does not imply that every reviewed method solves MPC explicitly. It states the control-theoretic interface a world model should support for planning: action-conditioned rollout, cost-relevant state exposure, constraint checking, uncertainty propagation, and repeated replanning.

Generative mechanisms represent uncertainty through different factorizations. An autoregressive world model typically decomposes a future sequence into ordered conditional distributions,

$$p_\theta(\mathbf{x}_{1:T} \mid \mathbf{c}) = \prod_{\tau=1}^{T} p_\theta(\mathbf{x}_\tau \mid \mathbf{x}_{<\tau}, \mathbf{c}), \tag{3}$$

where $\mathbf{x}_{1:T}$ may be visual tokens, BEV tokens, occupancy tokens, or trajectory tokens, and $\mathbf{c}$ denotes conditioning information. Autoregressive uncertainty is propagated through the

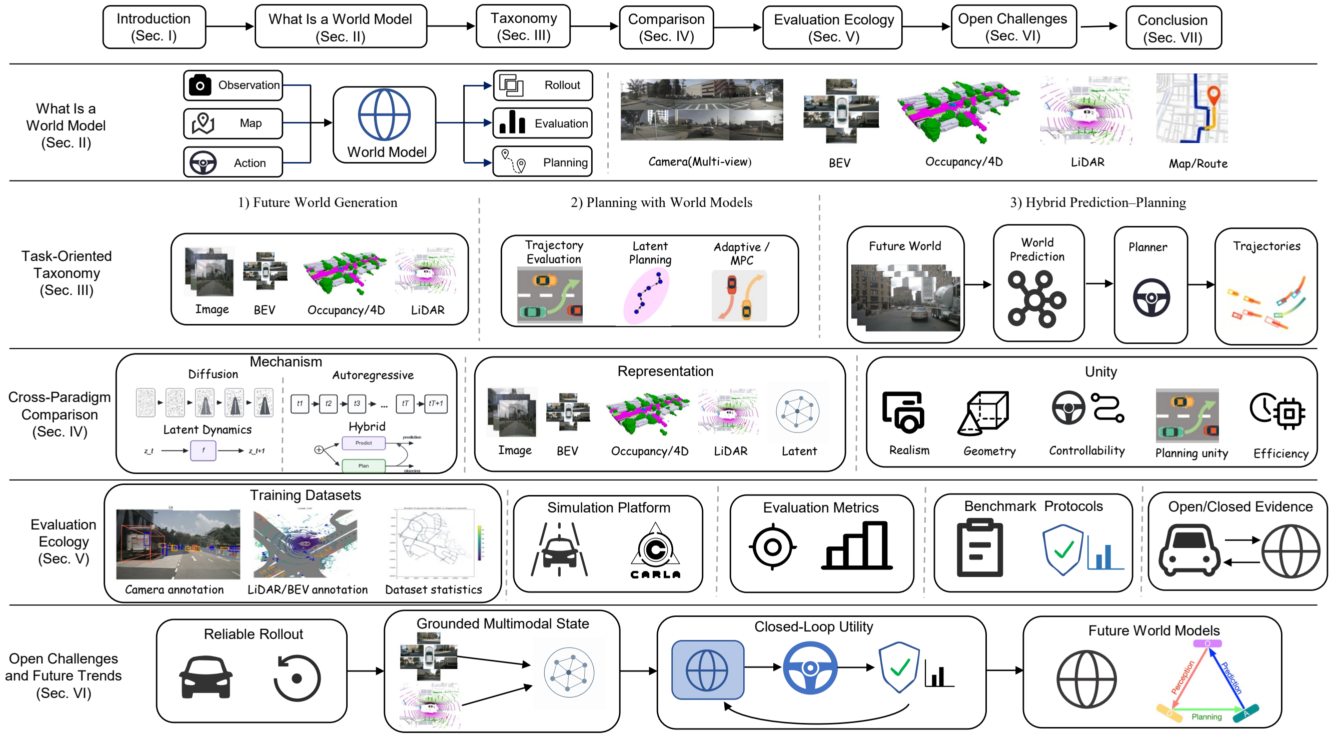

Fig. 2. Overall roadmap. We organize world models for auto-driving from the definition of the predictive transition interface to task-oriented taxonomy, cross-paradigm comparison, evaluation ecology, and future research challenges. The layout emphasizes the MPC-oriented perspective: generated futures should be assessed by their ability to support representation grounding, mechanism selection, planning utility, benchmark evidence, and closed-loop decision making.

generated history, which is flexible but can accumulate exposure error during long rollout. Diffusion-based world models instead define generation as a reverse denoising process,

$$p_\theta(\mathbf{x}_0 \mid \mathbf{c}) = \int p(\mathbf{x}_K) \prod_{k=1}^{K} p_\theta(\mathbf{x}_{k-1} \mid \mathbf{x}_k, \mathbf{c})\, d\mathbf{x}_{1:K}, \tag{4}$$

where uncertainty is induced by the noise path and reverse transition distribution. The diffusion formulation is powerful for multimodal high-dimensional generation, but multiple denoising steps and samples may be needed before uncertainty becomes actionable for a planner. Equations (1)–(4) provide the mathematical basis for later comparisons. The key question is whether the induced distribution can be queried, constrained, and converted into decision-relevant costs, beyond simply generating a plausible future.

C. Relation to Prediction, Simulation, and Planning

World models overlap with prediction, simulation, and planning, but are not identical to any one of these modules. Relative to motion prediction, they are more scene-level and state-oriented, covering trajectories as well as occupancy, BEV dynamics, and sensor observations. Relative to simulators, they model environment evolution, while simulators also include environment updates, agent behavior, sensor rendering, and evaluation interfaces. Relative to planners, they do not define the final control objective; they provide imagined futures, latent dynamics, or cost-relevant states.

The boundaries are nevertheless porous. Motion forecasting methods such as VectorNet, Scene Transformer, MTR, MotionLM, Wayformer, and occupancy flow fields provide ingredients for world modeling because they represent multi-agent futures, map context, and interaction uncertainty [19]–[24]. There is also a similar shift from trajectory extrapolation to scene-level future modeling through probabilistic tracking, structural recurrent interaction modeling, graph-based spatio-temporal prediction, hierarchical behavior prediction, and safety-critical trajectory prediction [25]–[29]. End-to-end planners and imitation-learning systems are included only when they expose intermediate BEV states, latent plans, or trajectory costs that clarify planner-usable rollout [30]–[32].

D. Scope and Comparison Dimensions

The survey covers three classes of work: *Future world generation models* generate future scenes or states in visual, BEV, occupancy, or LiDAR spaces. *Planning-oriented models* use imagined futures for trajectory scoring, policy learning, or MPC-style rollout. *Hybrid prediction-planning models* couple world modeling and decision making more tightly, sometimes as executable closed-loop testbeds. Pure perception, conventional trajectory prediction without world-state rollout, policy learning without a world representation, and language-only reasoning systems are not treated as primary world models.

We compare methods along six dimensions: task role, representation space, conditional inputs, modeling mechanism, output interface, and evaluation protocol. The dimensions

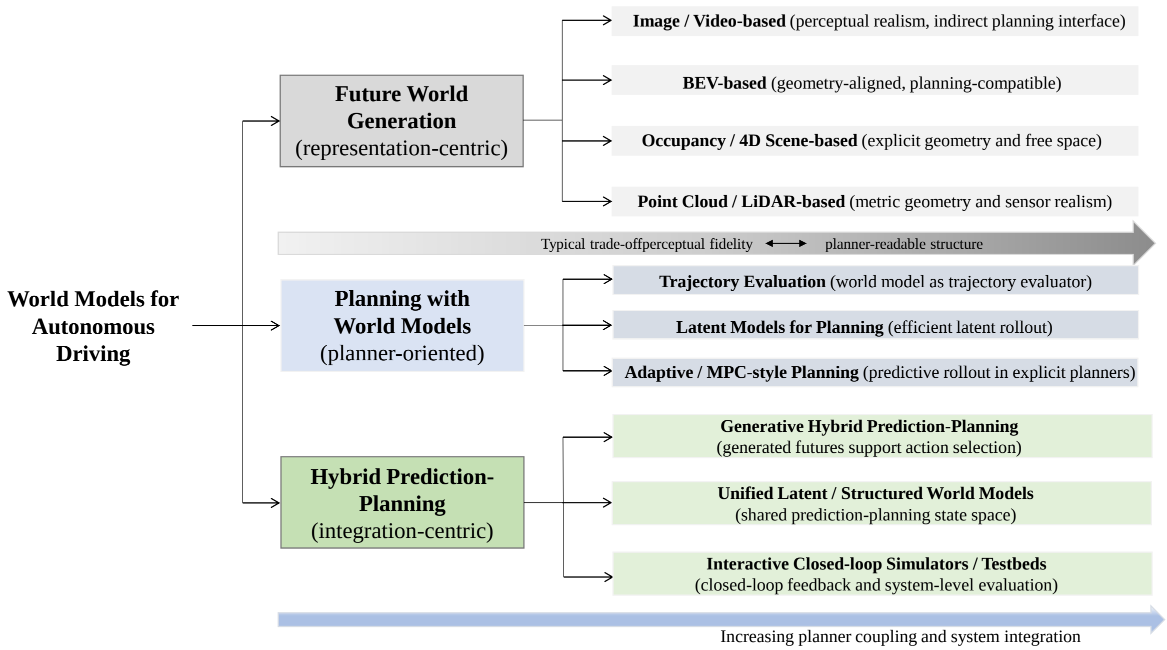

Fig. 3. A task-oriented taxonomy of autonomous driving world models. Existing methods can be organized into future world generation, planning with world models, and hybrid prediction-planning systems.

are needed because architecture alone does not reveal a model's contribution. A visually realistic video generator, a latent dynamics model, and a reactive simulator may all be called world models, but they support different parts of the autonomous driving stack. The task-role dimension identifies whether the model acts mainly as a generator, evaluator, planner component, or simulator component. The representation dimension specifies whether the state is visual, BEV, occupancy, LiDAR, object-centric, trajectory-based, or latent. The condition dimension describes how the model incorporates action, route, map, agent history, text, or user control. The mechanism dimension covers diffusion, autoregression, state-space modeling, transformers, graph neural networks, and hybrid designs. The output-interface and evaluation dimensions record how other modules consume the model and whether evidence comes from reconstruction, prediction, actionability, closed-loop driving, or safety testing.

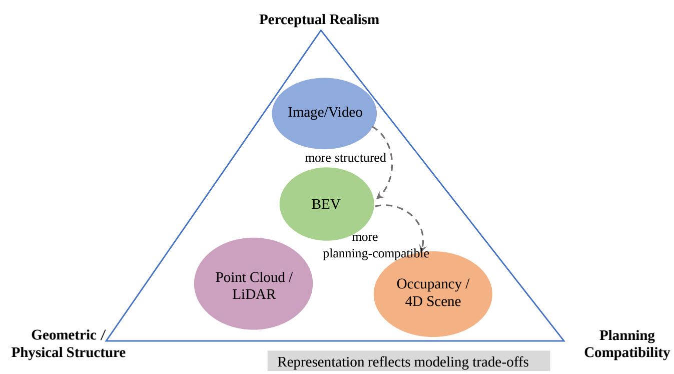

Fig. 4. Representation-space trade-offs for future world generation. Visual, BEV, occupancy, and LiDAR world models expose different balances among perceptual realism, geometric structure, sensor fidelity, and planning compatibility.

III. Task-oriented Taxonomy

Fig. 3 summarizes the task-oriented taxonomy for autonomous driving world models. The existing methods are first organized by task role, and representation choices and modeling mechanisms are then analyzed within each role.

Tables II–IV summarize representative methods under this taxonomy, emphasizing each method's approach, state/output interface, decision relevance, and evaluation evidence.

A. Future World Generation

Future world generation is the most direct entry point for driving world models. The central question is how the future environment should be represented and rolled forward. In driving, the generated future may be camera frames, BEV semantic maps, occupancy volumes, point clouds, trajectories, or scene graphs. Each space exposes different error modes: video exposes appearance and camera consistency, BEV exposes map alignment and agent layout, occupancy exposes false free space and collision geometry, and LiDAR exposes range realism and sensor sparsity. Thus, future generation must be interpreted with respect to the downstream task.

1) Visual World Models: Image- and video-based world models generate future observations directly in sensor space. Their advantage is perceptual interpretability for scenario visualization, data augmentation, failure reconstruction, and controllable rendering. Their limitation is that visual outputs are not automatically usable by a planner.

One line of work formulates visual world modeling as token-based sequence prediction. GAIA-1 encodes video, text,

TABLE II
REPRESENTATIVE AUTONOMOUS-DRIVING WORLD MODELS AND EVIDENCE (PART 1: FUTURE WORLD GENERATION)

Study	Approach	State / Output	Evidence and Planning Relevance
FIERY [33] (ICCV'21)	Probabilistic BEV latent dynamics with ego-motion conditioning.	BEV instance masks, motion fields, trajectories.	nuScenes/Lyft; scene rollout, mainly prediction.
GAIA-1 [34] (arXiv'23)	Discrete-token AR video WM with text and ego-action control.	Future camera video tokens.	Internal logs; controllable visual imagination.
DriveDreamer [35] (ECCV'24)	Conditional diffusion with HD maps, boxes, and ego actions.	Action video and traffic layout.	nuScenes; useful for generation/augmentation.
DriveDreamer-2 [36] (AAAI'25)	LLM layout control plus multi-view diffusion.	Text/map/layout-conditioned video.	nuScenes; stronger controllability, visual focus.
Copilot4D [37] (ICLR'24)	VQ tokenization and discrete diffusion for LiDAR sequences.	Future point-cloud tokens.	nuScenes/KITTI/AV2; geometry-preserving rollout.
OccWorld [38] (ECCV'24)	Occupancy tokenizer with spatio-temporal Transformer.	3-D occupancy and ego trajectory.	nuScenes; closer to planner-facing states.
DOME [39] (arXiv'24)	Spatio-temporal diffusion for long-horizon occupancy.	Multi-step occupancy states.	nuScenes; long-horizon geometric forecasting.
Drive-OccWorld [40] (AAAI'25)	BEV memory and action-conditioned occupancy decoding.	4-D occupancy, flow, cost cues.	nuScenes; directly supports candidate scoring.
RangeLDM [41] (ECCV'24)	Latent range-image diffusion with text/layout conditions.	LiDAR range-image sequences.	nuScenes; sensor-realistic simulation evidence.
SparseWorld [42] (AAAI'26)	Sparse dynamic-query modeling for efficient occupancy.	Sparse 4-D occupancy futures.	Occupancy forecasting; efficient dense rollout.

and action into a unified sequence, predicts future image tokens with an autoregressive transformer, and reconstructs high-resolution video with a diffusion decoder [34]. This separates high-level dynamics from rendering, but mainly supports controllable visual generation rather than structured planning.

Diffusion-based video generation provides another line. DriveDreamer conditions diffusion on HD maps, 3-D boxes, and text, while DriveDreamer-2 moves controllability upstream by inducing agent trajectories and scene layout from prompts [35], [36]. GAIA-2 extends this direction with a continuous video tokenizer, latent diffusion, multi-camera consistency, scene editing, inpainting, and rollout [43]. Recent work shifts from short-term appearance to long-horizon coherence: STAGE addresses train-test mismatch in streaming autoregressive generation, and Orbis studies continuous autoregressive flow matching with trajectory realism beyond FID or FVD [44], [45].

Visual world models are best understood as controllable future-synthesis models, useful when appearance, editing, and human inspection matter. Their weakness is the extra interface needed to convert visual futures into cost, risk, or action choices. They also face driving-specific challenges in multi-camera geometric coherence, long-horizon drift, and control fidelity: a video may look realistic while missing the intended lane change, acceleration, route, or interaction. Such failures motivate action-conditioned benchmarks such as ACT-Bench and WorldSimBench [17], [46]. For driving, the target is controllable visual simulation constrained by maps, boxes, trajectories, occupancy priors, and language rather than photorealism alone.

2) BEV World Models: BEV is a widely used intermediate representation because it aligns multi-view observations in a common ground-plane coordinate system. It preserves road topology, object layout, and motion relations while discarding appearance details weakly related to decisions. BEV is less photorealistic than video, but more compatible with detection, prediction, and planning.

Early work used BEV for future prediction. FIERY constructs BEV features from surround-view cameras, aligns history with ego motion, and predicts future instance segmentation and motion [33]. PowerBEV simplifies future instance association and shows that clear geometry can reduce heavy decoding [47]. Later methods make BEV more planner-facing: StretchBEV compresses BEV into a latent state and advances uncertainty through stochastic residual updates [48], CarFormer extracts slot-based object-centric BEV representations for scene dynamics and behavior [49], MILE couples BEV latent dynamics with policy learning and rollout [50], and BEVWorld uses action-conditioned BEV tokens for multimodal scene simulation [51]. This progression makes BEV a compact state where prediction, simulation, and planning meet.

BEV models sit between generation and planning. They balance geometry, compactness, interpretability, and utility, but lose vertical and fine 3-D structure needed for occlusion, free-space reasoning, and complex geometry. They inherit advances from BEV perception, where Lift-Splat-Shoot, BEVFormer, BEVDepth, and BEVFusion transform images, depth, temporal attention, and multi-sensor fusion into bird's-eye-view states [52]–[55]. World models add a temporal transition problem: the BEV state must remain coherent after ego motion, agent motion, and planned actions. BEV rollouts are inspectable through centers, velocities, lane relations, route corridors, dynamic occupancy, and cost maps, but may hide height, shape, traffic-light appearance, and subtle visual cues.

3) Occupancy-Centric 4-D Scene World Models: Occupancy and 4-D scene representations are closer to scene-level geometric reasoning. They model occupied space, dynamic evolution, and free space explicitly, making them naturally compatible with collision checking, free-space reasoning, and trajectory cost construction.

Diffusion-based occupancy generation is represented by DOME, which uses an Occ-VAE to compress occupancy into a continuous latent space and a spatio-temporal diffusion transformer for future prediction [39]. A second direction treats occupancy as a structured language of world states: OccLLaMA unifies occupancy, language, and action into a multimodal vocabulary, while Occ-LLM separates static and dynamic occupancy with a multi-scale VAE and improves long-sequence modeling with patch tokens and action prefusion [56], [57]. Drive-OccWorld links 4-D occupancy forecasting, action-controllable generation, and occupancy-based planning [40]; DriveDreamer4D uses generative world priors for novel-trajectory videos and 4-D scene reconstruction [58]; UNO and SparseWorld address efficiency and continuity with continuous occupancy fields and sparse dynamic queries [42], [59]. The common direction is to turn occupancy from geometric forecasting into a decision interface carrying action, language, uncertainty, and free-space information.

Occupancy-centric world models are among the most planner-friendly representations because safety depends on knowing which space is occupied, free, or uncertain. Dense fields express occluded regions, non-box-shaped objects, and multi-level geometry more naturally than 2-D boxes or trajectories, and support differentiable or rule-based collision checking. Their cost is expensive supervision, rollout, and evaluation; labels often require offline fusion or reconstruction, and occupancy alone does not encode intent. Recent occupancy-language-action and 4-D scene methods add action conditions, language tokens, and semantic priors, but their safety value still depends on how planners use uncertainty and rare-event information.

4) LiDAR World Models: Point-cloud and LiDAR world models emphasize metric geometry. Compared with image/video models, LiDAR captures 3-D structure and sensor geometry directly, but point clouds are sparse, unordered, and non-uniformly sampled, making generation and forecasting difficult.

RangeLDM projects LiDAR to range-view images and performs latent diffusion while preserving geometric structure [41]. OLiDM improves foreground object quality and controllability through object-scene progressive generation and semantic alignment [60]. Copilot4D tokenizes LiDAR observations into BEV tokens and predicts future point clouds with discrete diffusion and spatio-temporal transformers [37]. LidarDM generates latent 4-D worlds and uses stochastic raycasting to synthesize LiDAR observations [61]. LiDARCrafter converts text instructions into ego-centric scene graphs before generating boxes, trajectories, shape priors, and LiDAR sequences [62]. ViDAR uses future point-cloud prediction as geometric supervision for camera-only representation learning [63]. The shared emphasis is sensor-grounded geometry; the open question is how much this realism changes downstream behavior.

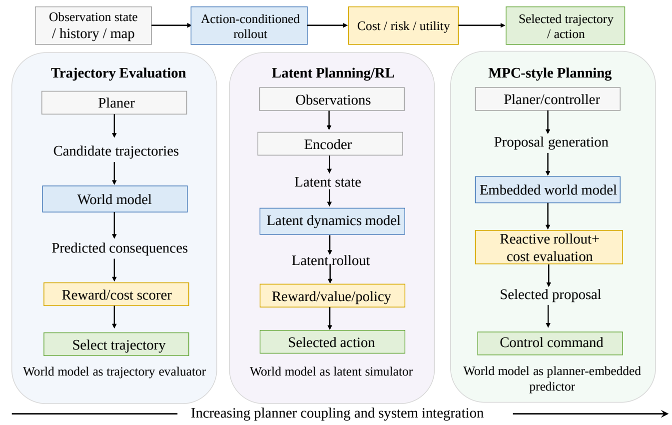

Fig. 5. Planning with world models. In this role, imagined futures are consumed by trajectory evaluation, latent policy learning, or MPC-style rollout rather than treated as standalone generated samples.

LiDAR world models are valuable when geometric fidelity and sensor consistency matter, but evaluation must account for object quality, temporal consistency, scan realism, and downstream utility. LiDAR generation is sparse, viewpoint-dependent, and governed by sensor physics: missing returns, range quantization, beam pattern, reflectance, boundaries, and occlusion affect whether generated scans can train or evaluate perception. Range-view diffusion, latent raycasting, and object-aware generation therefore encode sensor-sampling assumptions. LiDAR world models support data augmentation, closed-loop sensor simulation, and geometry-first validation, but behavior still depends on background-agent policies, map consistency, traffic rules, and route-conditioned ego feedback.

B. Planning with World Models

Planning with world models asks how imagined futures can improve decisions. The model shifts from future generator to decision-oriented predictive mechanism, whose output may be a reward, value, trajectory score, cost-relevant occupancy field, or latent transition. This changes the optimization target: a future generator can match logged data, whereas a planning world model must remain useful under counterfactual ego actions such as braking, lane changing, yielding, nudging, or following another route. Since logs provide one realized future, planning-oriented models must handle missing counterfactual supervision through candidate rollout, latent imagination, or MPC-style pipelines.

1) Trajectory Evaluation with World Models: Trajectory-evaluation methods first generate candidate trajectories and then use a world model to evaluate their consequences. MP3 and ST-P3 can be viewed as early forms of this idea because they combine future scene estimation with explicit trajectory scoring [64], [70]. WoTE makes the evaluator role clearer by mapping candidate trajectories to future BEV states and scoring them with a reward model [68]. Drive-OccWorld similarly

TABLE III
REPRESENTATIVE AUTONOMOUS-DRIVING WORLD MODELS AND EVIDENCE (PART 2: PLANNING WITH WORLD MODELS)

Study	**Key Findings**	**Planning Interface**	**Evaluation Scenario**
MP3 [64] (CVPR'21)	BEV-centric perception–prediction–planning.	BEV map, agent futures, ego trajectory.	Open/closed-loop CARLA-style evaluation.
MBIL [50] (NeurIPS'22)	Learned simulator plus imitation policy.	Latent rollout with policy loss.	Closed-loop urban simulation.
DTPP [65] (ICRA'24)	Differentiable trajectory trees and learned costs.	Agent futures and trajectory cost.	nuPlan-style open-loop planning.
Think2Drive [66] (ECCV'24)	Dreamer-style latent dynamics for efficient RL.	Latent transitions, rewards, policies.	Closed-loop CARLA-V2 tasks.
AdaptiveDriver [67] (ICRA'25)	Task-adaptive latent dynamics under shift.	Adapted rollout and policy.	Closed-loop CARLA tasks.
WoTE [68] (ICCV'25)	BEV WM for online trajectory evaluation.	Future BEV states and trajectory scores.	NAVSIM/Bench2Drive; MPC-aligned evidence.
Raw2Drive [69] (NeurIPS'25)	Raw-sensor WM aligned by privileged guidance.	Latent rollout for policy optimization.	CARLA v2; tests compact sensor latents.

predicts future occupancy and flow under observation-action conditions and uses them for occupancy-based planning [40].

The trajectory-evaluation paradigm is interpretable and easy to embed into proposal-based planners. A production-style planner can retain its candidate generator, rule checks, and comfort constraints while replacing or augmenting scoring. Unsafe candidates can be traced to predicted occupancy, distance to agents, rule violation, or learned reward. The weakness is coverage: if no candidate contains a safe maneuver, the evaluator cannot recover it; if the world model is poorly calibrated off distribution, risky actions may be scored too optimistically. TITS planning studies make the same point: trajectory quality depends on spatial feasibility, temporal evolution, and safety margins rather than endpoint accuracy alone [71], [72]. Candidate diversity, uncertainty estimation, and closed-loop validation are therefore central.

2) Latent World Models for Decision Learning: Latent world models for planning and reinforcement learning do not primarily reconstruct high-fidelity observations. They learn compact latent states where dynamics, reward, and value can be rolled out efficiently. PlaNet, Dreamer, DreamerV2, and DreamerV3 established the latent-imagination paradigm in general decision learning [73]–[76]. MuZero, EfficientZero, and TD-MPC show that latent models can also support search or trajectory optimization when predicting decision-relevant quantities [77]–[79].

In autonomous driving, Think2Drive uses Dreamer-style latent dynamics as a neural simulator for planner training [66]. PIWM moves from scene-level compression toward interaction-aware individual-level latent modeling [80]. Raw2Drive shows that learning reliable latent models from raw sensors is difficult and uses privileged inputs to guide raw-sensor learning [69]. AdaWM studies task and distribution shifts, showing that pretrained world models may require selective adaptation [81].

Latent world models are efficient and compatible with policy learning, but compact states may discard geometry, semantics, or interaction details essential for driving. Once observations are compressed into a recurrent state, many imagined rollouts can be evaluated without rendering full images or dense occupancy. General world-model agents such as World Models, PlaNet, Dreamer, DayDreamer, MuZero, EfficientZero, and TD-MPC demonstrate the value of latent dynamics in control and search [73], [74], [77]–[79], [82], [83]. Autonomous driving adds a stricter test: the latent state must preserve lane topology, traffic rules, small objects, occluded agents, and asymmetric interaction. TITS work on deep reinforcement learning and action-space design reaches a similar conclusion: learned policies remain effective only when the action abstraction and environment model respect driving constraints [84], [85]. Privileged supervision, structured auxiliary losses, or anchors to BEV, occupancy, or trajectory variables are often needed to prevent latent artifacts.

3) Adaptive World Models for MPC Planning: Adaptive and MPC-style methods embed predictive modeling into an explicit planner or controller stack. Instead of replacing the planner with a general latent simulator, they improve candidate generation, rollout, cost evaluation, and action selection.

The planning logic is continuous with classical planning and model predictive control. MPC optimizes predicted future behavior under dynamics, constraints, and cost functions over a receding horizon [12], [13], [86]. IEEE TITS planning studies similarly stress geometric feasibility, dynamic constraints, and traffic safety [71], [87]. Learning-based predictive control shows that the predictive model, safe set, value approximation, or terminal cost can be improved from data while retaining an optimization-based controller interface [79], [88]. A learned driving world model is therefore the predictive component that makes MPC-style reasoning richer, adaptive, or interaction-aware, not a replacement for MPC.

The Apollo EM planner and IDM provide examples of structured planning and interpretable longitudinal behavior models [89], [90]. PDM-C shows that open-loop trajectory fitting does not necessarily translate into closed-loop quality and that a classical centerline-search, IDM-proposal, internal-simulation, and scoring pipeline can remain strong [91]. AdaptiveDriver learns interpretable IDM parameters from ego history, neighboring agents, and lane-graph context, making

TABLE IV
REPRESENTATIVE AUTONOMOUS-DRIVING WORLD MODELS AND EVIDENCE (PART 3: HYBRID PREDICTION–PLANNING SYSTEMS)

Study	Key Findings	Decision Interface	Evaluation Scenario
Drive-WM [92] (CVPR'24)	Multi-view imagination for action comparison.	Future video and reward cues.	nuScenes-style planning evidence.
GenAD [93] (ECCV'24)	Instance-centric generative end-to-end driving.	Scene-token futures and ego plans.	nuScenes planning evaluation.
DrivingGPT [94] (ICCV'25)	Unified AR modeling of observations and actions.	Visual/action token rollout.	nuPlan/NAVSIM planning evidence.
World4Drive [95] (ICCV'25)	Intention-conditioned physical latent futures.	Latent futures for trajectory selection.	nuScenes/NAVSIM, closed-loop-oriented.
DriveArena [96] (ICCV'25)	Generative closed-loop traffic scene evolution.	Reactive simulation and reward feedback.	Closed-loop simulator evidence.
DriveLaW [97] (arXiv'25)	Shared latents for planning and video.	Plan-conditioned generation/action alignment.	Planning benchmark evidence.
UniDrive-WM [98] (arXiv'26)	Unified language, vision, and action tokens.	Driving tokens for reasoning and rollout.	Unified-task evaluation; VLA-oriented.

rollout more reactive and behavior-aware [67]. DTPP lies near this category because it jointly optimizes prediction and planning through trajectory trees and learned costs [65].

The MPC-style approach is close to deployable planning stacks and remains interpretable. Its limitation is that richer interaction modeling increases complexity and may still be constrained by the planner design.

MPC-style world modeling also makes uncertainty operational. In a planner, uncertainty changes braking distance, lateral clearance, fallback behavior, and interaction timing. A world model used in MPC should produce uncertainty that cost functions can consume, such as probabilistic occupancy, multi-modal agent motion, stochastic latent transitions, or ensemble rollouts. TITS studies on probabilistic prediction and risk-aware trajectory prediction support the same view: uncertainty is most useful when tied to interaction risk, safety-critical scenarios, and decision consequences rather than reported only as distributional fit [25], [29].

The historical link to classical planning matters. IDM, lane-graph search, and rule-based proposal generation are not obsolete because simple structured priors encode traffic conventions and constraints. A practical direction is to let learned world models handle complex interaction and adaptation while retaining classical priors for safety envelopes, fallback behavior, and interpretable costs.

C. Hybrid Prediction-Planning World Models

Hybrid prediction-planning methods couple future modeling and action selection more deeply. They move from using a world model as an auxiliary predictor toward a system component for joint prediction, planning, and evaluation. This reduces the mismatch between what the model predicts and what the planner needs, because the representation can be optimized for both future consistency and action utility. The cost is harder evaluation: failures may come from representation collapse, poor generation, incorrect cost learning, weak exploration, or planner instability. Hybrid papers therefore need stronger ablations and clearer protocols than single-task predictors.

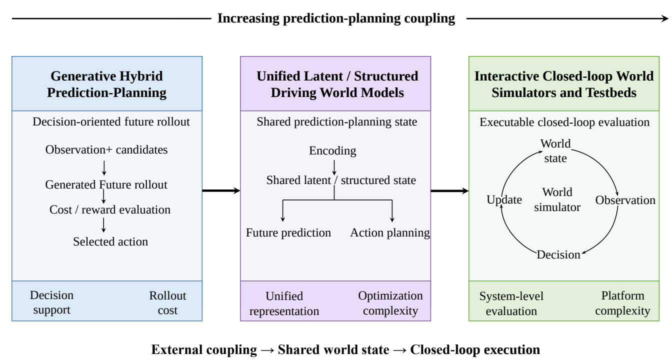

Fig. 6. Hybrid prediction-planning world models. The family moves from external coupling between future generation and decision scoring toward shared latent states and executable closed-loop simulators.

1) Generative Hybrid Prediction-Planning: Generative hybrid methods use generated futures directly in action selection. Drive-WM rolls out multiple candidate trajectories with a multi-view video world model and compares imagined futures with image-based reward and tree search [92]. ReSim improves the reliability of generated future evaluation by adding real driving data, non-expert simulation data, and a Video2Reward module [99]. DrivingGPT unifies image tokens and action tokens in an autoregressive model, jointly predicting future observations and planning trajectories [94].

Generative hybrid methods show that observation-level world modeling can move toward planning. This route is useful for consequences hard to express with hand-crafted costs: image-level futures can reveal weather, lighting, object appearance, lane markings, and unusual visual contexts, while learned rewards or perception modules assess candidate behavior. The challenge is that observation-space rollout is expensive and fragile because video artifacts may affect downstream perception and realistic-looking futures may still violate geometry or interaction constraints. These limitations motivate combinations with structured BEV, occupancy, or object-level states.

2) Unified Latent Driving World Models: Unified latent and structured models try to make prediction and planning share the same internal state. GenAD organizes surrounding-agent evolution and ego trajectories in an instance-centric structured latent space [93]. World4Drive introduces driving intention and selects among latent futures under different intention conditions [95]. WorldDrive transfers vision-motion representations to downstream planning through representation inheritance and future-latent distillation [100]. DriveLaW uses mid-level latents from a video generator as conditions for action generation [97]. UniDrive-WM and DriveWorld-VLA point toward a broader reasoning-action-generation framework [98], [101].

The main gain is a weaker boundary between prediction and planning. The cost is higher optimization and evaluation complexity because generation quality, latent consistency, control stability, and safety are no longer independent. Unified latent models also raise a representation-governance question: which variables should be shared between prediction and control, and which should remain task-specific? Fully shared latents encourage compactness but can entangle visual details with action choice; partially shared representations preserve planner-relevant variables while allowing generation-specific decoders. Instance-centric, intention-aware, and VLA-style models offer different answers, but all still require geometric grounding and closed-loop tests.

3) Closed-loop Interactive World Simulators: Interactive world simulators move world modeling to the system level. DriveArena combines a traffic manager with a generative world module so that traffic evolution, scene update, and multi-view observation generation form a closed loop [96]. DrivingSphere uses a structured 4-D occupancy world for dynamic environment composition and visual scene synthesis [102]. RealEngine integrates scene reconstruction, traffic participant composition, camera/LiDAR rendering, and agent assessment in a unified testbed [103]. UniSim is an important precursor for neural sensor simulation, scene editing, and closed-loop evaluation [104].

The value of interactive simulators extends beyond observation synthesis: they provide executable interaction, feedback, and evaluation. Their weakness is that credibility depends on state construction, rendering, background-agent behavior, protocol design, and the world model itself. Interactive simulators shift the question from logged-future prediction to credible intervention after the ego vehicle changes the scene. Credible intervention requires reactive background agents, route-consistent map updates, sensor rendering, collision handling, traffic-light state, and metrics. Data-driven simulators such as Waymax, ScenarioNet, Nocturne, and SimNet show how logged scenes can seed multi-agent rollout, while neural rendering and generative world modules improve observation realism [105]–[108]. World-model simulators should therefore report intervention robustness, scenario coverage, agent reactiveness, and failure reproducibility in addition to visual quality and average driving score.

IV. Cross-Paradigm Comparison

Motivated by the receding-horizon logic of MPC, this section compares methods through a control-oriented question: *what does a world model provide to a planner that repeatedly predicts, scores, constrains, and updates ego actions?*

A. Mechanism–Task Matching

Modeling mechanisms should be compared by task role, not ranked as generators. Diffusion suits high-dimensional future generation when controllability and perceptual quality matter [39], [43], [109]. Autoregressive transformers provide token interfaces for video, text, action, occupancy, and object-centric BEV states [34], [49], [56], [110]. Latent dynamics and RSSM-style models fit rollout, value estimation, and policy improvement because repeated imagined rollouts are cheap [66], [74], [79]. Hybrid designs connect prediction and planning, but shift difficulty into multi-objective optimization and evaluation.

Each mechanism has a different control profile. Diffusion represents multi-modal futures, but sampling is costly and constraints are hard to maintain. Autoregressive models tokenize flexibly, but sequential rollout can accumulate exposure bias. State-space and recurrent latent models are efficient for imagination, but compact states are hard to verify. Graph and object-centric models preserve relations and maps, but may under-represent appearance and dense geometry. A practical rule is to choose the simplest representation that preserves downstream variables: visual diffusion for scenario visualization, BEV or occupancy for trajectory scoring, compact latents for policy learning, and hybrid systems for closed-loop simulation. This avoids judging all world models by photorealism alone.

B. Cross-Representation Trade-offs

Visual representations support realism and simulation, but require conversion before planning. BEV balances geometry and compactness, but loses vertical structure. Occupancy and 4-D scenes are planner-friendly, but expensive to train and evaluate. LiDAR preserves metric geometry and sensor realism, but is sparse and difficult to generate. Latent states enable efficient rollout, but are less interpretable and harder to validate. In MPC terms, representations differ in how directly they expose optimizer states, constraints, and costs.

The trade-off also depends on supervision and uncertainty. Images are abundant but weakly structured; BEV and occupancy labels often require 3-D detection, map alignment, fusion, or annotation; LiDAR provides geometry but not semantics or intent. Latent states can be learned from reward or reconstruction, but may fail silently if supervision misses safety-critical variables. Uncertainty must also be actionable: samples, cell-level probabilities, trajectory modes, or stochastic latents should imply slowing down, yielding, widening margins, or fallback behavior. The MPC lens asks whether uncertainty changes the feasible set, safety margin, terminal condition, or candidate cost.

C. Utility-Oriented Comparison

Generation quality and driving utility are not equivalent. FID and FVD measure distributional realism, but do not prove correct geometry, control following, or planning improvement [111], [112]. From the MPC perspective, utility appears when a model improves prediction, candidate rollout, cost evaluation, constraint satisfaction, or closed-loop replanning. Planning-oriented methods such as WoTE, AdaptiveDriver, and AdaWM are less visually direct but more tied to trajectory selection, risk-aware planning, and adaptation [67], [68], [81]. Bench2Drive shows that open-loop metrics often miss real driving ability, while NAVSIM provides a middle ground between log replay and fully reactive simulation [15], [16]. World-model utility should therefore be evaluated through the full development and validation loop.

Utility-oriented comparison should ask what downstream decision changes, which failure mode becomes easier to detect or avoid, how sensitive the benefit is to distribution shift, horizon length, and candidate quality, and whether failures are inspectable. In MPC-style systems, the model must also be fast enough for repeated receding-horizon rollouts and translate uncertainty into constraints or costs. These questions are stricter than image similarity or trajectory error, but closer to autonomous-driving evidence.

V. Datasets, Simulators, Evaluation, and Benchmarks

A. Training Datasets

This subsection treats datasets as training and supervision resources rather than as evaluation protocols. Video-centric datasets such as DrivingDojo support photorealistic future synthesis, language-conditioned prediction, and action-conditioned generation [113]. Multimodal datasets such as KITTI, nuScenes, Waymo, and nuPlan provide calibrated camera, LiDAR, radar, map, pose, and trajectory information for BEV, fusion, planning, and map-conditioned modeling [14], [114]–[116]. Occupancy datasets such as Occ3D, OpenOccupancy, Cam4DOcc, and UniOcc provide dense 3-D or 4-D supervision for geometry-aware world modeling [117]–[120]. LiDAR-rich resources remain important for point-cloud forecasting and sensor-consistent simulation. The practical issue is coverage matching: datasets must supply the modalities, annotations, and temporal context required by the target representation, not only more samples.

Data requirements differ across representation spaces. Video models need weather, lighting, traffic, camera, and ego-motion diversity; BEV and occupancy models need calibration, ego pose, synchronization, maps, and geometric labels; planning models need routes and expert trajectories; LiDAR models need temporally aligned scans and sensor-consistent geometry. Motion forecasting datasets such as Argoverse, Argoverse 2, Waymo Open Motion, and Lyft Level 5 remain important because they capture interaction and map-conditioned futures [121]–[124]. Long-tail coverage remains difficult, so useful datasets should document scenario distribution, route diversity, interaction difficulty, annotation quality, and representation-specific supervision, not size alone.

B. Simulation Platforms and Ecosystems

Simulation platforms determine how models interact with environments. CARLA provides configurable sensors, conditions, and closed-loop execution [125]. MetaDrive and LimSim emphasize scalable scenario construction and traffic simulation [126], [127]. Waymax uses real driving logs to initialize multi-agent scenes and provides reactive agents, route-aware states, and closed-loop metrics [105]. Learned ecosystems combine behavior simulation with observation generation, turning world models from offline predictors into simulator components.

Classical and learned simulators have complementary strengths. Classical simulators provide controllable physics, repeatable execution, and explicit rules, but often lack photorealism and traffic diversity. Learned simulators inherit real-log statistics and richer observations, but may be unreliable under interventions far from the data manifold. Hybrid simulators combine explicit traffic structure with learned rendering or behavior. For world-model research, simulators should expose state, route, intervention interface, metric API, visual output, and background-agent assumptions.

C. Evaluation Metrics

Evaluation must follow output type. Image and video models are often evaluated by FID, FVD, PSNR, SSIM, or human preference, but these metrics mainly measure visual similarity [111], [112]. Occupancy and point-cloud models require IoU, mIoU, Chamfer distance, temporal consistency, and object quality. Action- or instruction-conditioned models require controllability and behavior alignment, while planning-oriented models require safety, progress, comfort, rule compliance, and closed-loop success. Trajectory-prediction studies show that static displacement errors can be weakly correlated with driving performance [128]. TITS work on crash mitigation, spatial-temporal planning, and risk-aware prediction further suggests reporting safety-relevant margins under constrained and interactive conditions, rather than average prediction error alone [29], [71], [72].

Metrics should be layered. Perceptual metrics ask whether observations resemble real data; geometric metrics ask whether states are spatially correct; controllability metrics ask whether conditions are followed; planning metrics ask whether ego behavior improves; safety metrics ask whether rare severe failures are reduced. Reporting only visual quality cannot establish planning utility, while reporting only closed-loop score may hide causality. Evaluation should also separate average quality from tail risk: a model that improves average FVD or displacement error may still miss rare cut-ins, occluded pedestrians, emergency vehicles, or construction zones. Tail-sensitive metrics, scenario categories, and failure analysis are necessary complements.

D. Benchmark Protocols

Unlike training datasets, benchmark protocols define the evaluation contract under which models are compared. In autonomous driving, the same resource may support both learning and benchmarking, but the analytical function differs: datasets provide learnable experience and supervision, whereas protocols specify task formulation, splits, allowed inputs, simulator assumptions, metrics, reporting rules, and comparison settings. Benchmarks therefore connect data, simulator, metric, and comparison rules. nuPlan is relevant here through its planning protocol, simulator, reactive agents, and planning metrics [14]. Bench2Drive improves closed-loop comparability through standardized routes, interaction categories, and decomposed driving skills [15]. WorldSimBench targets perceptual quality and actionability by asking whether generated videos support decision tasks [17]. Shared protocols are essential because world models have heterogeneous outputs and many papers use private settings.

The protocol determines what progress is visible. Log replay emphasizes representation quality, imitation, or prediction under fixed futures; non-reactive closed loop measures ego stability while limiting interaction evidence; reactive simulation evaluates planning improvement, interaction robustness, and closed-loop safety but depends on background-agent realism; scenario-generation benchmarks test controllability and rare-event coverage but may not measure everyday driving. Future benchmarks should state their evidence target clearly: representation quality, actionability, planning improvement, interaction robustness, safety validation, or scenario-generation capability.

E. Open-Loop and Closed-Loop Evidence

Open-loop evaluation is cheap, scalable, and reproducible, but measures agreement with logged futures under fixed distributions. Once a world model enters a planner, its predictions affect ego actions and future observations, creating a dynamics gap between open-loop fit and closed-loop performance. Closed-loop evaluation tests safety, progress, comfort, and interaction more directly, but depends on simulator fidelity, background-agent realism, sensor rendering, and protocol design. Balanced evaluation should combine open-loop debugging, counterfactual controllability tests, and closed-loop system tests, treating the two settings as layered evidence rather than a valid/invalid dichotomy.

VI. Open Challenges and Future Trends

A. Long-Horizon Reliability

Long-horizon reliability remains a bottleneck. The issue is whether imagined futures stay stable enough for safe decisions, not merely whether a model can generate longer videos or occupancy sequences. Rollout errors can accumulate into temporal drift, trajectory deviation, and degraded safety margins, and non-reactive agents can amplify early errors. NAVSIM and BridgeSim suggest that open-loop and closed-loop performance correlate less as the horizon grows because of objective mismatch and compounding execution error [16], [18]. Future protocols should jointly cover rollout length, planner feedback, multi-agent interaction, uncertainty calibration, and safety constraints. Hierarchical world models may connect sub-second collision avoidance, multi-second interaction, and route-level progress, but only if they preserve uncertainty across horizons.

Fig. 7. MPC-oriented future challenges for autonomous-driving world models. The figure summarizes five connected directions: reliable long-horizon rollout, unified multimodal state representation, bridges from generation quality to planning utility and closed-loop evidence, efficient scalable deployment, and foundation-model-assisted safety validation.

B. Unified Multimodal World Representation

Driving requires visual appearance, 3-D geometry, maps, interaction relations, actions, and sometimes language. A unified representation should provide a state abstraction robust to sensor type, sensitive to geometry and action, and useful for prediction and planning. Shared latents, BEV latents, occupancy latents, cross-modal token alignment, and multimodal pretraining move in this direction, but feature-level alignment does not guarantee geometric, causal, or control consistency. Language and VLA models add semantic reasoning while introducing hallucination, latency, grounding, and verification issues [11], [129], [130].

Recent embodied multimedia research broadens static sensor fusion into embodied multimodal signal processing. Zuo et al. organize embodied multimedia around data acquisition, communication, cognition, and evaluation layers that support a perception–decision–action feedback loop [131]. For autonomous-driving world models, this matters because camera, LiDAR, maps, language, trajectory commands, and interaction cues are task-dependent signals whose value depends on preserving physical semantics and improving future action. Robust representations should make invariances explicit: appearance may change, but road topology and collision geometry should remain stable; intent may be uncertain, but kinematics should constrain futures; language can describe scenarios, but physical feasibility should filter them. DriveLM and DriveVLM illustrate language-grounded reasoning for driving decisions, while Dilu represents knowledge-driven LLM-based driving [132]–[134].

C. Bridging Generation, Planning, and Evaluation

The gap between generation and planning is structural. A model can generate realistic futures and still fail to improve decisions, because perceptual fidelity, temporal consistency, and conditional following do not automatically become safety, progress, comfort, or interaction stability. BridgeSim attributes the open-loop to closed-loop gap to objective mismatch and compounding execution error, while WorldSimBench separates perceptual evaluation from actionability [17], [18]. Future benchmarks should build an evidence chain across perceptual quality, behavior alignment, and closed-loop consequence. Decision-aware training can help by preserving variables that change planner decisions, including free space, collision risk, agent intent, rule violation, comfort, and route progress.

D. Efficiency and Scalability

Efficiency is not a secondary engineering issue. World models must support online inference, repeated candidate rollout, and sometimes closed-loop simulation. Dense video, occupancy, point clouds, diffusion sampling, long token sequences, and large language modules all increase latency and memory. Compact latents reduce rollout cost but may lose structure. Useful directions include structured latents, distillation, quantization, parameter-efficient tuning, token reduction, sparse activation, hierarchical inference, and event-triggered computation. Scalability should also be measured at the system level: one efficient rollout may still be too slow for hundreds of candidates, while a slightly less accurate model may be more useful if it supports fast uncertainty-aware rollout. Future papers should report inference frequency, rollouts, hardware, memory, and integration assumptions.

E. Foundation Models and Multimodal Decision Making

Foundation models extend world modeling from future synthesis toward language-grounded reasoning, scenario construction, and multimodal decision making. Contrastive vision-language pretraining, represented by CLIP, shows how natural-language supervision can produce transferable visual-semantic features [135]. In autonomous driving, semantic alignment is useful only when grounded in traffic geometry, agent interaction, and feasible motion: recognizing yield, cut-in, or construction-zone events matters only if they modify trajectories, margins, costs, or route-level decisions.

One practical direction is to use VLMs as semantic interfaces around world models rather than replacements for physical modeling. DriveLM formulates driving reasoning as graph visual question answering, while DriveVLM couples vision-language understanding with hierarchical planning [132], [133]. Such systems bridge high-level instructions and low-level trajectory planning: language and visual concepts can condition multi-view attention, vectorized scene tokens, BEV or occupancy states, and candidate rollout evaluation [32]. The world model provides the predictive state interface, while the VLM supplies semantic priors, task intent, and open-vocabulary reasoning. This view also reframes safety validation: testing and scenario-generation studies address rare and safety-critical cases [136]–[138]. World-model safety work separately stresses the need to verify feasibility, safety relevance, and planning utility in grounded and closed-loop settings [139].

VII. Conclusion

In this survey, we review autonomous-driving-specific world models from the model predictive control perspective. Future world generation asks how the driving world can be represented and rolled forward; planning-oriented models ask how imagined futures become scores, rewards, values, or costs; and hybrid models move toward unified representations and executable closed-loop systems. Across these categories, representation and mechanism choices trade off realism, geometry, planning compatibility, controllability, and efficiency. World models should not be evaluated only as generators: a realistic future is useful only when a downstream system can consume it in a stable, interpretable, and verifiable way. The field needs stronger links between generation objectives, planning interfaces, evaluation protocols, long-horizon reliability, multimodal state abstractions, scalable simulation, and safety-oriented benchmarks.

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